

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271069

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

MCCARRA; Sidney

FLAPPER CHECK VALVE ASSEMBLY

Abstract

A swing check valve includes a body and a swing assembly, the body including an inner volume having a middle portion extending between an upstream end portion and a downstream end portion and a wall that surrounds the inner volume and the swing assembly including a disc having an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position and a valve seat having an outer circumferential surface in contact with the wall. The body also includes a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path.

Inventors: MCCARRA; Sidney (League City, TX)

Applicant: SCV Valve, LLC (Santa Fe, TX)

Family ID: 1000007708954

Assignee: SCV Valve, LLC (Santa Fe, TX)

Appl. No.: 18/585476

Filed: February 23, 2024

Publication Classification

Int. Cl.: F16K15/03 (20060101)

U.S. Cl.:

CPC F16K15/03 (20130101);

Background/Summary

TECHNICAL FIELD

[0001] Aspects of the present disclosure relate generally to valves and valve systems useful, for example, for fluid handling. In particular, aspects of the present disclosure relate to methods and systems for flapper check valves.

BACKGROUND

[0002] Valve devices, such as flapper check valve assemblies, are useful for fluid handling in, for example, the oil and gas, power, chemical, water works, waste water, and manufacturing industries. Flapper check valves are robust systems that are operable to prevent backflow in fluid handling systems, including those operated in harsh environments and under severe conditions. In order to perform in these environments, valve systems include components that are resistant to debris, corrosion, and wear.

[0003] Despite this robust construction, while the valve is in service, components of the valve assembly require regular inspections and maintenance. Even when inspections and maintenance are performed regularly, wear accumulates due to the operation of moving parts of the valve, affecting locations that cannot be accessed without disconnecting the valve from adjacent fluid handling components. For example, a pipeline inspection device, which can be referred to as a “pig,” can cause damage or wear to critical components of the check valve, including the seat. In order to service or replace these components, it can be necessary to disconnect the valve from upstream and downstream components. However, removing the valve from a line is a time consuming process that introduces significant expense.

[0004] Flapper check valves are sometimes designed for use with an external component to help the valve function. For example, to provide a level of protection for the flapper during pigging, a check valve may include an external lock lever. Alternatively, a flapper check valve may be equipped with a slam dampener to help reduce the water hammer effect. While useful, these external components are typically integrated with the flapper check valve and thus are not designed to be changed.

SUMMARY

[0005] In one aspect, a swing check valve may include a body, the body including an inner volume having a middle portion extending between an upstream end portion and a downstream end portion and a wall that surrounds the inner volume. The body may also include a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path and a swing assembly including a disc including an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position and a valve seat having an outer circumferential surface in contact with the wall.

[0006] In another aspect, a valve system includes a body, the body including an inner volume having a middle portion extending between an upstream end portion and a downstream end portion, a wall that surrounds the inner volume, and a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path. The body may also include a key assembly positioned on an exterior surface of the wall, the key assembly including a hollow hub with a body and a keyway, wherein the keyway is a cutout on an inner surface of the hollow hub and a shaft with a body and a keyseat, wherein the keyseat is a cutout on an outer surface of the shaft, the hollow hub and the shaft being concentric.

[0007] In yet another aspect, a method of assembling a swing check valve system may include forming a body including an inner volume having a middle portion extending between an upstream end portion and a downstream end portion, a wall that surrounds the inner volume and a flow path extending from the upstream end portion to the downstream end portion, and a valve seat having an outer circumferential surface in contact with the wall, wherein the middle portion includes the flow

path and an upper recess positioned above the flow path. The method may further include placing a swing assembly within the body including a disc including an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position and the upstream surface including a protrusion that protrudes towards the upstream end portion when the disc is in the closed position such that the protrusion extends furthest at a radial center portion of the disc, and wherein the protrusion protrudes past a downstream end of the valve seat in an upstream direction.

[0008] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the disclosed embodiments, as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate various exemplary embodiments and together with the description, serve to explain the principles of the disclosed embodiments.

[0010] FIG. 1 is a side view of a flapper check valve assembly according to aspects of the present disclosure.

[0011] FIG. 2A is a cross-sectional view of the flapper check valve system in a closed position, according to aspects of the present disclosure.

[0012] FIG. 2B is a cross-sectional view of the flapper check valve system in an open position, according to aspects of the present disclosure.

[0013] FIG. 3 is an enlarged view of Detail A of FIG. 2A.

[0014] FIG. 4A is a front view of the key assembly, according to aspects of the present disclosure.

[0015] FIG. 4B is a cross-sectional side view of the key assembly, according to aspects of the present disclosure.

[0016] FIG. 5A is a side view of a valve system with a lock lever, according to aspects of the present disclosure.

[0017] FIG. 5B is a side view of a valve system with a lock gear, according to aspects of the present disclosure.

[0018] FIG. 5C is a side view of a valve system with a slam dampener, according to aspects of the present disclosure.

DETAILED DESCRIPTION

[0019] Both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the features, as claimed. As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. Moreover, in this disclosure, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of $\pm 10\%$ in the stated value.

[0020] FIG. 1 is a side view of a flapper check valve assembly **100** according to aspects of the present disclosure. Flapper check valve assembly **100**, also referred to herein as flapper check valve system **100**, may include a body **102**, a key assembly **104**, and a swing assembly or flapper assembly **206** (shown in FIGS. 2A-2B).

[0021] Body **102** may include a pair of opposed ends formed by annular flanges or ports, including an upstream port **105** and a downstream port **106**, ports **105** and **106** being part of an upstream end portion **108** and a downstream end portion **110**, respectively. Body **102** may include a middle

portion **114** with a bonnet **116**. Body **102** is formed of a corrosion-resistant material, such as a metal material. Body **102** may be formed of carbon steel, for example.

[0022] Ports **105** and **106** may be sized and shaped for connection to upstream and downstream components, and may include a series of bolt holes (not shown) to allow ports **105** and **106** to facilitate leak-free connections to these components.

[0023] FIGS. 2A-2B are cross-sectional views of flapper valve assembly **100** of FIG. 1, with FIG. 2A showing a closed position of flapper assembly **206** and FIG. 2B showing an open position of flapper assembly **206**. With reference to FIGS. 2A-2B, an inner volume **218** of body **102** may house swing assembly **206**. Key assembly **104** (FIG. 1) may be positioned through an exterior surface **112** and through the wall **200** of body **102**.

[0024] Inner volume **218** of body **102** is surrounded by wall **200**, with an open portion of body **102** being closed with a bonnet **116**. Inner volume **218** may include a flow path **204** with a diameter which flows through the flapper check valve assembly **100** from upstream end portion **108** to downstream end portion **110**. Inner volume **218** at middle portion **114** may further include an upper recess **202** positioned above flow path **204** and below bonnet **116**.

[0025] A swing assembly **206** may form a one-way valve that permits flow of fluid in a flow direction from upstream end portion **108** toward downstream end portion **110**, and prohibits flow of fluid in a direction from downstream end portion **110** toward upstream end portion **108**. Swing assembly **206** may include a disc **208**, an arm **210**, and a shaft **212**. Flapper assembly **206** may further contact a removable valve seat **300**. Swing assembly **206** may be moveable between a closed position in which disc **208** rests on removable valve seat **300**, as shown in FIG. 2A, and an open position in which disc **208** is spaced away from and does not contact removable valve seat **300**, as shown in FIG. 2B.

[0026] Disc **208** may have a diameter that is larger than the diameter of flow path **204** adjacent to upstream end portion **110**. Disc **208** may be shaped to block the entirety of flow path **204** when in a closed position, fluidly separating upstream end portion **108** from downstream end portion **110**. Disc **208** may include an upstream surface **214** that faces upstream end portion **108** when flapper assembly **206** is in the closed position and a downstream surface **216** that faces downstream end portion **110** when flapper assembly **206** is in the closed position. Disc **208** may be flat or may include protrusions that create a curved surface.

[0027] In some embodiments, upstream surface **214** may be flat or curved and include one or more bumps protruding towards upstream end portion **108**. In other embodiments, upstream surface **214** is entirely smooth. In some aspects, disc **208** may be circular because flapper check valve assembly **100** has a circular flow path. As shown, disc **208** may be substantially dome-shaped. For example, upstream surface **214** may protrude towards upstream end portion **108** in such a way that upstream surface **214** extends furthest at a radial mid-portion, or in particular, the radial midpoint, of disc **208**, which forms a contact point **220** for engaging a pig **222** (represented by a dashed-line box in FIG. 2B) within flow path **204** during pigging. "Pigging" as used herein refers to a method of cleaning pipelines in which a device sized to the diameter of the flow path of the pipeline (known as a pig) is inserted into the pipeline and moves from the upstream end to the downstream end and cleans and/or inspects the interior of the pipeline as it progresses.

[0028] In some embodiments, a center portion of upstream surface **214**, and therefore the portion of upstream surface **214** that protrudes the furthest towards upstream end portion **108**, contact point **220**, extends past a downstream end of valve seat **300**. In particular, upstream surface **214** may extend beyond a midpoint of removable valve seat **300**, as measured in an axial direction defined by valve seat **300** and as shown in FIG. 2A. If desired, upstream surface **214** may extend beyond an upstream end of valve seat **300**, protruding farther upstream than valve seat **300**, as measured axially. Upstream surface **214** may be entirely curved or may have flat ends near where upstream surface **214** contacts removable valve seat **300**, as shown in FIG. 2A. For example, upstream surface **214** may include an outer circumference configured to sealingly contact a downstream axial

end of removable valve seat **300**. The outer periphery of upstream surface **214** may include a tapered surface or chamfer for contacting removable valve seat **300**.

[0029] Arm **210** may connect to downstream surface **216** of disc **208** in any suitable manner. For example, arm **210** may be connected by a pin or other fastener to allow disc **208** and arm **210** to be disengaged from one another. Arm **210** may additionally couple to shaft **212** so that shaft **212** rotates with arm **210** and moves with disc **208**. Arm **210** may be coupled to shaft **212** so there is no independent rotational movement between arm **210** and shaft **212**.

[0030] Shaft **212** may be positioned in upper recess **202**. As shown in FIG. 2A, shaft **212** may extend through inner volume **218** in a direction perpendicular to flow path **204** and into and/or through two portions of wall **200**. For example, in FIGS. 2A-2B, flow path **204** extends from left to right while shaft **212** extends into and out of the sheet. Shaft **212** may have an exterior portion outside of body **102** and an interior portion. In some embodiments, shaft **212** may only be within body **102** (e.g., an entirety of shaft **212** is located between front and rear sides of body **102** that are located orthogonal to upstream and downstream end portions **108** and **110**). The interior portion of shaft **212** may extend through key assembly **104**, through upper recess **202** (in a perpendicular direction to flow path **204** as described above), and into wall **200**. Shaft **212** may be supported by one or more bearings to reduce friction when shaft **212** is rotating.

[0031] In the exemplary embodiment shown in FIG. 2B, shaft **212** is positioned such that disc **208** and arm **210** may abut wall **200** when in the open position. In some embodiments, wall **200** may include a wall portion **205** that contacts swing assembly **206** in the open position. Additionally, a stopper (not shown) may protrude from the wall portion **205** instead to contact swing assembly **206** in the open position instead. In other embodiments, swing assembly **206** may have enough clearance to swing open and not contact wall **200**.

[0032] When sufficient force presses against upstream surface **214** (e.g., a sufficient flow in a downstream direction), the flow from upstream end portion **108** to downstream end portion **110** causes the swing assembly **206** to automatically swing into the open position. For example, when fluid (e.g., oil, other petroleum products, water, etc.) introduced from upstream end portion **108** applies sufficient force on upstream surface **214**, disc **208** may swing (upwards and towards the downstream end portion **110** in FIG. 2A) so as to allow fluid communication in a direction from upstream end portion **108** toward downstream end portion **110**. Conversely, swing assembly **206** protects the system from backflow as gravity and force against downstream surface **216** automatically pivots swing assembly **206** into the closed position. In other words, swing assembly **206** may automatically move into the closed position and block reverse or upstream flow of fluid in a direction from downstream end portion **110** toward upstream end portion **108**. Shaft **212** may act as a pivot point for swing assembly **206**. Bearings may be located at each end of shaft **212**, thus allowing shaft **212** to rotate when disc **208** rotates.

[0033] In some embodiments, shaft **212** may terminate within wall **200** on one side while extending through wall **200** on the other side, allowing for external components to connect to shaft **212** and interact with swing assembly **206**. Arm **210** may act as a pivot arm by coupling disc **208** to shaft **212**. Because the upstream surface **214** may be substantially dome shaped, during pigging, pig **222** contacts disc **208** at contact point **220** prior to contacting any part of swing assembly **206**, and prior to contacting any part of removable valve seat **300**. Advantageously, the shape of disc **208** protects removable valve seat **300** from wear or damage during pigging. During pigging, pig **222** may “ride” along upstream surface **214** while pig **222** is passing through valve assembly **100**. This may further protect the disc **208** (and other parts of assembly **100**) from damage during the pigging process.

[0034] Removable valve seat **300** may be a ring-shaped member inserted within a recess of body **102**. Removable valve seat **300** may be formed of a corrosion and wear-resistant material that is suitable for repeated contact with the angled or chamfered surface of disc **208**. Removable valve seat **300** may also include an inclined or chamfered surface. Removable valve seat **300** may be

formed of a metal material, such as a material including carbon steel. In some aspects, removable valve seat **300** may include a polymeric material, such as polytetrafluoroethylene (PTFE), polyether ether ketone (PEEK), other polymers, and composites thereof. As mentioned above, removable valve seat **300** may be coaxially disposed with respect to disc **208** when disc **208** is in the closed position to contact an outer circumference of upstream surface **214** such that a portion of upstream surface **214** protrudes at least partially through an inner diameter of removable valve seat **300**.

[0035] In FIG. 3, removable valve seat **300** may be secured to wall **200** by a set screw **302**. While one set screw **302** is shown in FIG. 3, there may be more than one set screw **302** used to fasten valve seat **300**. Seals **304** and **306** (e.g., O-rings) may provide additional sealing for removable valve seat **300**. Set screw **302** allows removable valve seat **300** to be removed and replaced when needed, therefore extending the life of flapper check valve assembly **100**. Removable valve seat **300** may be firmly secured to body **102** without the need to be permanently attached to body **102** (e.g., by welding or by being cast or otherwise formed with body **102**). The assembly of removable valve seat **300** may aid in reducing or eliminating vibration and extend the life of flapper check valve assembly **100** by reducing wear on swing assembly **206** and removable valve seat **300**.

[0036] Removable valve seat **300** may include an outer circumferential surface **308** that is in contact with wall **200**. Outer circumferential surface **308** may face wall **200** so as to be free of welding, or other mechanisms for permanently fixing removable valve seat **300** to wall **200**. In some aspects, removable valve seat **300** may be fixed by the force applied by swing assembly **206**. Thus, removable valve seat **300** may be secured so as to abut wall **200** in a manner that allows removal of removable valve seat **300** without the need to use cutting tools to separate removable valve seat **300** from wall **200**.

[0037] Key assembly **104**, formed on exterior surface **112** of flapper check valve assembly **100** in FIG. 1, may be used to connect different accessory devices to flapper check valve assembly **100**. As shown in FIG. 4A and FIG. 4B, key assembly **104** may include a hollow hub **400** with a body **402** and a keyway **404**, a shaft **212** with a body **408** and a keyseat **410**, and a key **412**. Hollow hub **400** may also have an outer surface **414** and an inner surface **416** while shaft **212** has an outer surface **418**. Hollow hub **400** and shaft **212** may be concentric so that hollow hub **400** surrounds at least a portion of shaft **212**. For example, shaft **212** may fit within hollow hub **400** such that inner surface **416** and outer surface **418** are in contact with one another.

[0038] Keyway **404** may be a cutout on inner surface **416** extending (e.g., in an arc) along at least a portion of body **402**. Keyseat **410** may be a cutout on outer surface **418** extending along at least a portion of body **408**. Key **412** may be a metal piece shaped to fit within keyseat **410** and to move within keyway **404** when keyseat **410** and keyway **404** are overlapped. Key **412** may take different shapes; for example, key **412** may be substantially half-moon shape with a flat, straight side and a curved side, or key **412** may be substantially rectangular with all straight sides. Since the bottom portion of key **412** sits in keyseat **410** and the top portion of key **412** fits in keyway **404**, keyseat **410** may resemble the bottom half of key **412** while keyway **404** may resemble the top half of key **412**. Thus, keyseat **410** and keyway **404** resemble the shape of key **412** when overlapped.

[0039] Inserting key **412** into keyseat **410** and keyway **404** of key assembly **104** prevents the independent motion of hollow hub **400** and shaft **212** and ensures both rotate together. When key **412** is engaged, hollow hub **400** may be blocked from rotation around shaft **212** and, instead, is permitted to rotate with shaft **212**. As can be seen in FIG. 4A, the geometric constraints of key assembly **104** restricts the manual movement of hollow hub **400** and shaft **212** when key **412** is engaged.

[0040] Key assembly **104** may be limited to 90 degrees of rotation. For example, key assembly **104** may be at rest at a first position **420** and can be rotated 90 degrees counterclockwise to a second position **422**. In FIG. 4A first position **420** of key **412** is represented with solid lines, with a second position **422** being represented by dashed lines. Key assembly **104** may be similarly capable of

being rotated **90** degrees clockwise from second position **422** to first position **420**.

[0041] In aspects of the present disclosure, hollow hub **400** may be formed by a portion of an external component (e.g., a lock lever **500**, a lock gear **502**, or a slam dampener **504**, shown in FIGS. 5A-5C, respectively, and described below). Hollow hub **400** may be configured to interact with swing assembly **206** via shaft **212**. When hollow hub **400** is a portion of an external component, hollow hub **400** is positioned on the exterior portion of shaft **212**. As described above, when key **412** is not engaged hollow hub **400** can move independently of shaft **212**. This may allow for easy interchangeability as the external component can be swapped when key **412** is not engaged. When key **412** is engaged, however, movements of the external component can cause shaft **212** to rotate and affect the position of disc **208**. While three exemplary components are described herein, external components that are configured to interact with swing assembly **206** are within the scope of this disclosure.

[0042] Key assembly **104** allows for numerous parts to interact with swing assembly **206** in an interchangeable manner, including lock lever **500**, lock gear **502**, and slam dampener **504**, shown in FIGS. 5A-5C respectively. Each of lock lever **500**, lock gear **502**, and slam dampener **504** may act as an extension to shaft **212** by connecting and securing to key assembly **104**.

[0043] As shown in FIG. 5A, lock lever **500** may include a lever **506** that can move between a first position and a second position. When key **412** is inserted within key assembly **104**, swing assembly **206** may rotate with lock lever **500** and allows lock lever **500** to control the position of swing assembly **206**. Lock lever **500** may operatively control swing assembly **206** to be in a closed position in which disc **208** rests upon removable valve seat **300** and closes flow path **204** when key **412** is in first position **420** and to be in an open position in which disc **208** is spaced away from removable valve seat **300** and opens flow path **204** when key **412** is in second position **422**. Lock lever **500** may additionally include a lock pin **508** to manually lock the lock lever **500** in second position **422** which may be desired for pigging and/or inspection.

[0044] Lock gear **502** may include a gear **510** that can twist between a first position and a second position. Because lock gear **502** may be connected to shaft **212**, when key **412** is inserted swing assembly **206** rotates with lock gear **502** and allows lock gear **502** to control the position of swing assembly **206**. Lock gear **502** may operatively control swing assembly **206** to be in a closed position in which disc **208** rests upon removable valve seat **300** and closes flow path **204** when key **412** is in first position **420** and to be in an open position in which disc **208** is spaced away from removable valve seat **300** and opens flow path **204** when key **412** is in second position **422**. Similarly to lock lever **500**, lock gear **502** allows manual opening of flow path **204** which may be desired for pigging and/or inspection.

[0045] Slam dampener **504** may include a hydraulic assembly **512** with an adjustable governor to regulate the speed at which disc **208** closes. In general, slam dampener **504** may be configured to decelerate disc **208** during closing to prevent slamming against removable valve seat **300**. Slowing the close speed of swing assembly **206** may help reduce a water hammer effect, thus minimizing sudden upstream pressure to avoid damage to removable valve seat **300**, disc **208**, wall **200**, and other internal components of flapper check valve assembly **100**.

[0046] It will be apparent to those skilled in the art that modifications may be made in the disclosed systems and methods without departing from the scope of the disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and embodiments be considered as exemplary only.

Claims

1. A swing check valve, comprising: a body including: an inner volume having a middle portion extending between an upstream end portion and a downstream end portion; a wall that surrounds

the inner volume; and a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path; a valve seat having an outer circumferential surface in contact with the wall; and a swing assembly including a disc having an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position; wherein the upstream surface of the disc includes a protrusion that protrudes towards the upstream end portion of the inner volume when the disc is in the closed position such that the protrusion extends furthest at a radial center portion of the disc, and wherein the protrusion protrudes past a downstream end of the valve seat in an upstream direction.

2. The swing check valve of claim 1, wherein the valve seat is removable.

3. The swing check valve of claim 1, further comprising a stopper that contacts the downstream surface when the disc is in an open position.

4. The swing check valve of claim 1, wherein the valve seat has an inclined or chamfered sealing surface.

5. A valve system, comprising: a body including: an inner volume having a middle portion extending between an upstream end portion and a downstream end portion; a wall that surrounds the inner volume; and a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path; and a key assembly positioned on an exterior surface of the wall, the key assembly including: a hollow hub with a body and a keyway, wherein the keyway is a cutout on an inner surface of the hollow hub; and a shaft with a body and a keyseat, wherein the keyseat is a cutout on an outer surface of the shaft, wherein the hollow hub and the shaft are concentric.

6. The valve system of claim 5, the key assembly further including a key configured to fit within the keyseat and to extend past the outer surface of the shaft in a radial direction.

7. The valve system of claim 5, wherein at least a portion of the keyseat of the shaft is circumferentially surrounded by the inner surface of the hollow hub.

8. The valve system of claim 5, wherein the hollow hub is configured to receive a key that connects the hollow hub with the shaft by contacting the keyway and the keyseat such that the key prevents rotation of the hollow hub relative to the shaft.

9. The valve system of claim 5, further comprising: a valve seat having an outer circumferential surface in contact with the wall; the shaft with an exterior portion outside of the body of the valve system and an interior portion, the interior portion extending through the key assembly, through the upper recess in a direction perpendicular to the flow path, and into the wall; and a swing assembly including: a disc having an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position; and an arm connected to the shaft and to the downstream surface such that the arm allows the disc to rotate with the shaft; wherein the hollow hub is positioned on the exterior portion of the shaft, wherein the upstream surface of the disc includes a protrusion that protrudes towards the upstream end portion when the disc is in the closed position such that the protrusion extends furthest at a radial center portion of the disc, and wherein the protrusion protrudes past a downstream end of the valve seat in an upstream direction.

10. The valve system of claim 9, wherein the hollow hub is a portion of a lock lever, the lock lever including a lever that can move between a first position and a second position, wherein the lock lever operatively controls the swing assembly to be in a closed position in which the disc rests upon the valve seat and closes the flow path when in the first position, and wherein the lock lever operatively controls the swing assembly to be in an open position in which the disc is spaced away from the valve seat and opens the flow path when in the second position.

11. The valve system of claim 10, the lock lever further including a lock pin to manually lock the lock lever in the second position.

- 12.** The valve system of claim 10, wherein the lock lever moves from the first position to the second position by the key assembly being rotated, via the lock lever, **90** degrees.
- 13.** The valve system of claim 9, wherein the hollow hub is a portion of a lock gear, the lock gear configured to move between a first position and a second position, wherein the lock gear operatively controls the swing assembly to be in a closed position in which the disc rests upon the valve seat and closes the flow path when in the first position, and wherein the lock gear operatively controls the swing assembly to be in an open position in which the disc is spaced away from the valve seat and opens the flow path when in the second position.
- 14.** The valve system of claim 13, wherein the lock gear moves from the first position to the second position by the key assembly being rotated, via the lock gear, **90** degrees counterclockwise about an arcuate path, and wherein the lock gear moves from the second position to the first position by rotating 90 degrees clockwise about the arcuate path.
- 15.** The valve system of claim 9, wherein the hollow hub is a portion of a slam dampener, the slam dampener including a hydraulic assembly, and wherein the slam dampener is configured to decelerate the disc during closing.
- 16.** The valve system of claim 15, the slam dampener further including an adjustable governor configured to regulate the speed at which the disc decelerates.
- 17.** A method of assembling a swing check valve system, the method comprising: forming a body including: an inner volume having a middle portion extending between an upstream end portion and a downstream end portion; a wall that surrounds the inner volume; and a flow path extending from the upstream end portion to the downstream end portion, wherein the middle portion includes the flow path and an upper recess positioned above the flow path; placing a swing assembly within the body including a disc having an upstream surface that faces the upstream end portion when the disc is in a closed position and a downstream surface that faces the downstream end portion when the disc is in the closed position; wherein the body houses a valve seat having an outer circumferential surface in contact with the wall, wherein the upstream surface includes a protrusion that protrudes towards the upstream end portion when the disc is in the closed position such that the protrusion extends furthest at a radial center portion of the disc, and wherein the protrusion protrudes past a downstream end of the valve seat in an upstream direction.
- 18.** The method of claim 17, wherein the valve seat is removable.
- 19.** The method of claim 17, further including providing a stopper that contacts the downstream surface when the swing assembly is in an open position.
- 20.** The method of claim 17, wherein the valve seat has a sealing surface that is inclined or chamfered.
-